

# FINGERPRINT DOOR UNLOCK SYSTEM Using Arduino

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This simple fingerprint sensor project using Arduino can be very useful for door security, forensics, crime investigation, personal identification, attendance system and much more. In the future, there could be many more applications like fingerprint based driving licenses, bank accounts operation and so on. The whole system works under a simple algorithm called matching algorithm, which is used to compare previously-stored templates of fingerprints against users' fingerprints for authentication purposes.

A key is normally used for traditional door opening, but it provides

very poor security. In this project, only when an authorised person places a finger on the sensor, the door unlocks and the LCD displays a welcome message along with that person's name.

## Circuit and working

The circuit shown in Fig. 1 operates using a 12V power supply. An Arduino microcontroller (MCU) requires only 5V but the solenoid electric lock requires 12V. As Arduino Uno has an inbuilt 5V voltage regulator, a common 12V supply can be used for the whole system.

The brain of the circuit is Ar-

duino Uno MCU board (BOARD1). It is based on ATmega328P/ATmega328P and has 14 digital input/output (I/O) pins, six analogue inputs, 32k flash memory, 16MHz crystal oscillator, a USB connection, power jack, ICSP header and reset button, among others. It can be programmed using Arduino IDE software.

Fingerprint sensor module R305 (connected across CON2) has UART interface with direct connections to the MCU or to the PC through max232/USB serial adaptor. The user can store fingerprint data in the

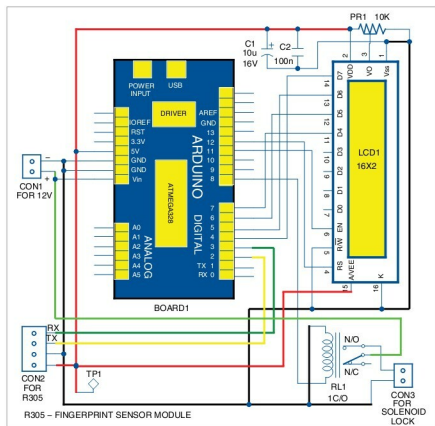


Fig. 1: Circuit diagram of the fingerprint door unlock system

## Test Points

| Test point | Details |
|------------|---------|
| TP1        | 5V      |

(Voltage measure w.r.t. GND)

## PARTS LIST

|                                              |                             |
|----------------------------------------------|-----------------------------|
| <b>Semiconductors:</b>                       |                             |
| BOARD1                                       | - Arduino Uno               |
| <b>Resistors (all 1/4-watt, ±5% carbon):</b> |                             |
| PR1                                          | - 10-kilo-ohm preset        |
| <b>Capacitors:</b>                           |                             |
| C1                                           | - 10µF, 16V electrolytic    |
| C2                                           | - 100nF, ceramic disk       |
| <b>Miscellaneous:</b>                        |                             |
| RL1                                          | - 5V, 1C/O relay            |
| LCD1                                         | - 16x2 alphanumeric display |
| CON1                                         | - 2-pin connector           |
| CON2                                         | - DC supply                 |
| CON3                                         | - 4-pin connector           |
| CON3                                         | - R305                      |
| CON3                                         | - 2-pin connector           |
| CON3                                         | - Solenoid lock             |



Fig. 2: Initial state



Fig. 3: Valid finger



Fig. 4: Invalid finger